

More exercises

Write Lewis structures (incl. formal charges) and predict structures using the VSEPR model for:

- a) IF_5
- b) SOCl_2
- c) ClO_3^-
- d) SF_4
- e) CH_3OH
- f) XeF_2
- g) NO_2^-

Strategy:

- 1) Write skeletal structure of molecule
- 2) Count the total number of valence electrons
- 3) Draw all bonds as single bonds and fill remaining electrons to complete octets
- 4) If it does not add up, use lone-pairs to generate double or triple bonds.

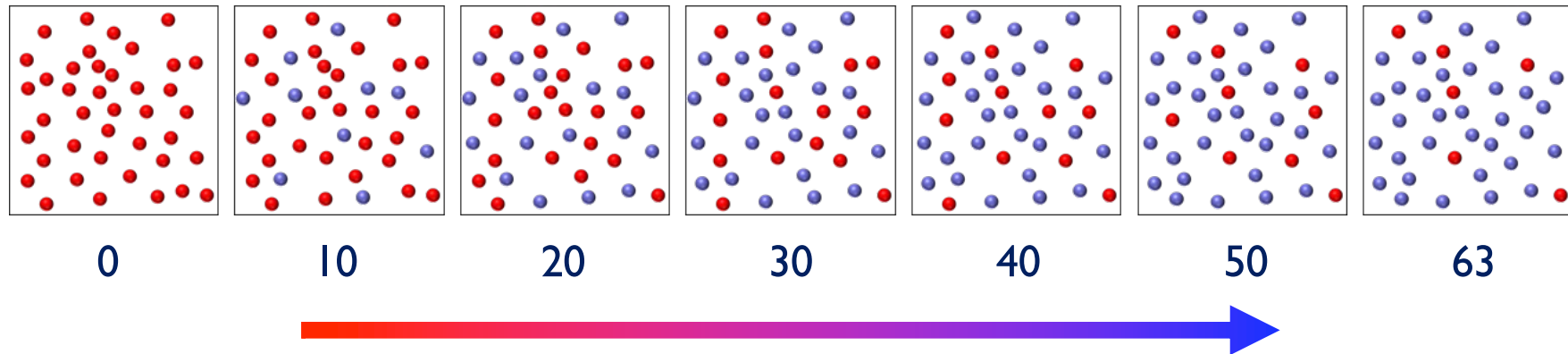
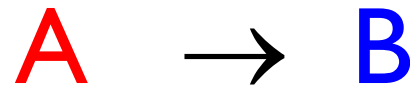
Lecture 7: Chemical Kinetics and Equilibria (Ch. 12+13)

26021 Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

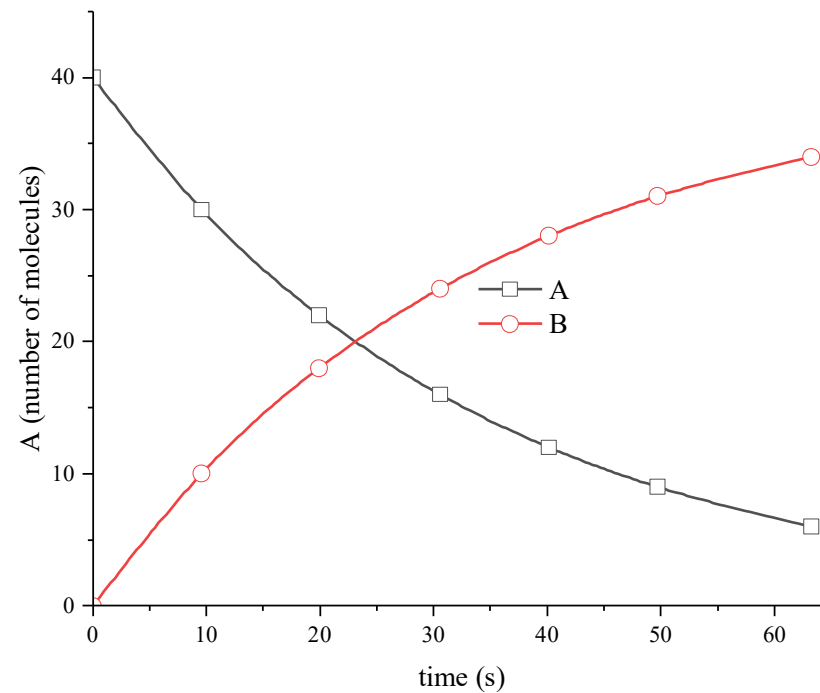
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Reaction rates



$$r = -\frac{d[A]}{dt} = \frac{d[B]}{dt}$$

$$r \approx -\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$$



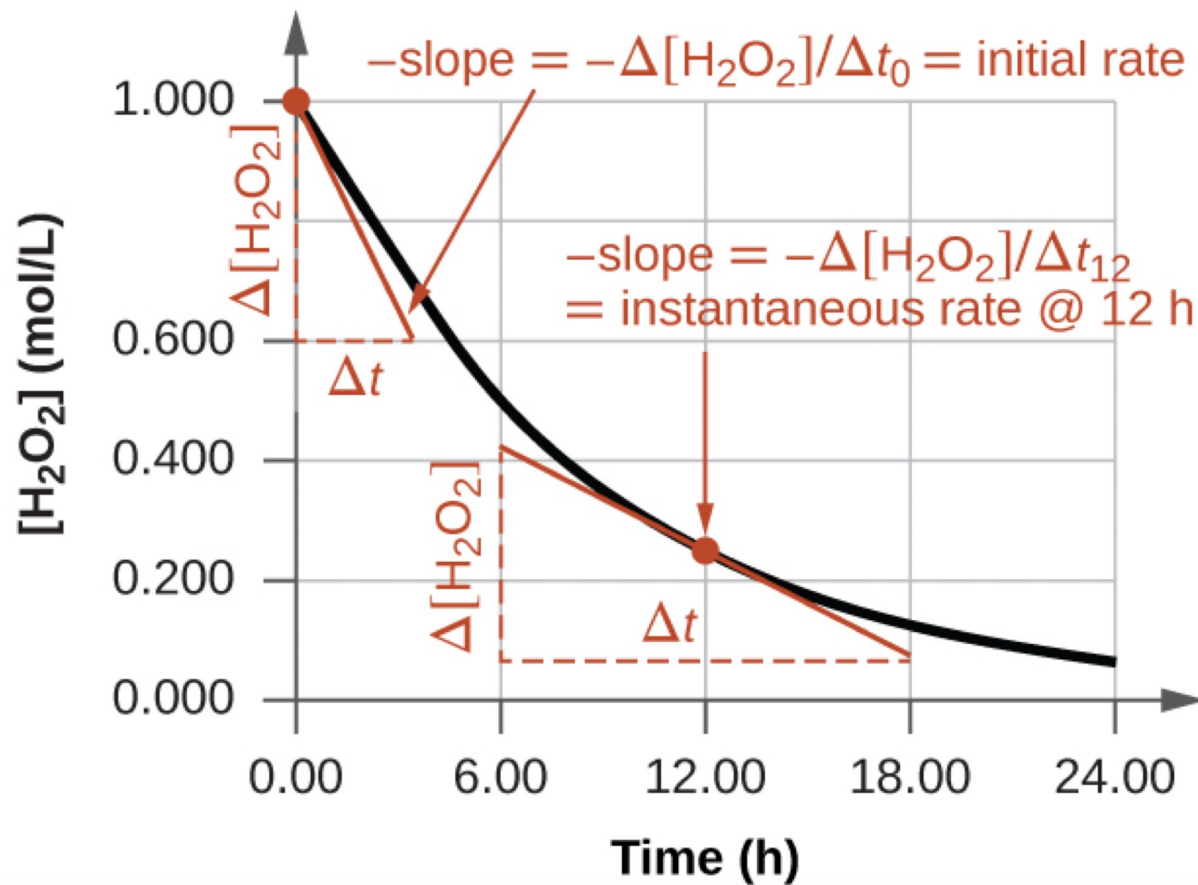
Reaction rates - example



$$\begin{aligned} \text{rate of decomposition of H}_2\text{O}_2 &= - \frac{\text{change in concentration of reactant}}{\text{time interval}} \\ &= - \frac{[\text{H}_2\text{O}_2]_{t_2} - [\text{H}_2\text{O}_2]_{t_1}}{t_2 - t_1} \\ &= - \frac{\Delta[\text{H}_2\text{O}_2]}{\Delta t} \end{aligned}$$

Time (h)	[H ₂ O ₂] (mol L ⁻¹)	Δ[H ₂ O ₂] (mol L ⁻¹)	Δt (h)	Rate of Decomposition, (mol L ⁻¹ h ⁻¹)
0.00	1.000	-0.500	6.00	0.0833
6.00	0.500			
12.00	0.250	-0.250	6.00	0.0417
18.00	0.125			
24.00	0.0625	-0.062	6.00	0.010

Reaction rates - example



The slope of a line tangent to the curve at any point is the instantaneous rate at that time.

$$r \approx -\Delta[\text{H}_2\text{O}_2]/\Delta t$$

versus

$$r = -d[\text{H}_2\text{O}_2]/dt$$

Reaction rates and stoichiometry



$$r = -\text{d}[\text{H}_2]/\text{d}t = -\text{d}[\text{I}_2]/\text{d}t = \frac{1}{2} \text{d}[\text{HI}]/\text{d}t$$

To generalize, for the reaction



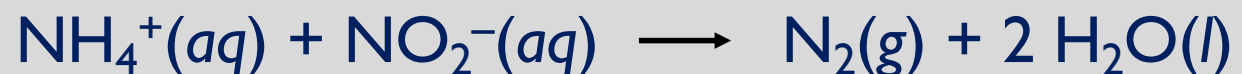
The rate becomes:

$$r = -a^{-1} \text{d}[\text{A}]/\text{d}t = -b^{-1} \text{d}[\text{B}]/\text{d}t = c^{-1} \text{d}[\text{C}]/\text{d}t = d^{-1} \text{d}[\text{D}]/\text{d}t$$

Reactants (decrease)

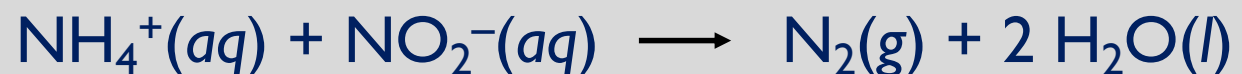
Products (increase)

Concentration and rate - Example



Experiment Number	Initial NH_4^+ Concentration (M)	Initial NO_2^- Concentration (M)	Observed Initial Rate (M/s)
1	0.0100	0.200	5.4×10^{-7}
2	0.0200	0.200	10.8×10^{-7}
3	0.0400	0.200	21.5×10^{-7}
4	0.0600	0.200	32.3×10^{-7}
5	0.200	0.0202	10.8×10^{-7}
6	0.200	0.0404	21.6×10^{-7}
7	0.200	0.0606	32.4×10^{-7}
8	0.200	0.0808	43.3×10^{-7}

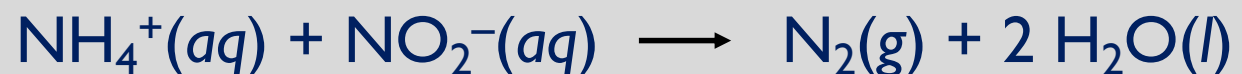
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When $[\text{NH}_4^+]$ doubles, the initial rate doubles

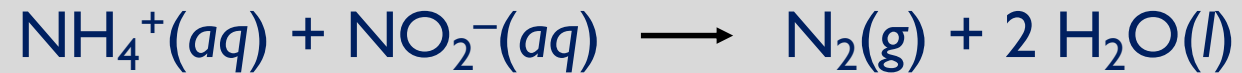
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Likewise, when $[\text{NO}_2^-]$ doubles, the initial rate doubles

Concentration and rate - Example



$$r \propto [\text{NH}_4^+] \quad \text{and} \quad r \propto [\text{NO}_2^-]$$

which gives the *rate law*:

$$r = k [\text{NH}_4^+] [\text{NO}_2^-]$$

where k is the *rate constant*.

k is experimentally determined, unique for a given reaction and changes with temperature (see later)

Note: for gaseous species, we use partial pressure instead of concentration



Rate laws

$$r = k [\text{NH}_4^+] [\text{NO}_2^-]$$

Exponents tell the order of the reaction with respect to each reactant.

This reaction is

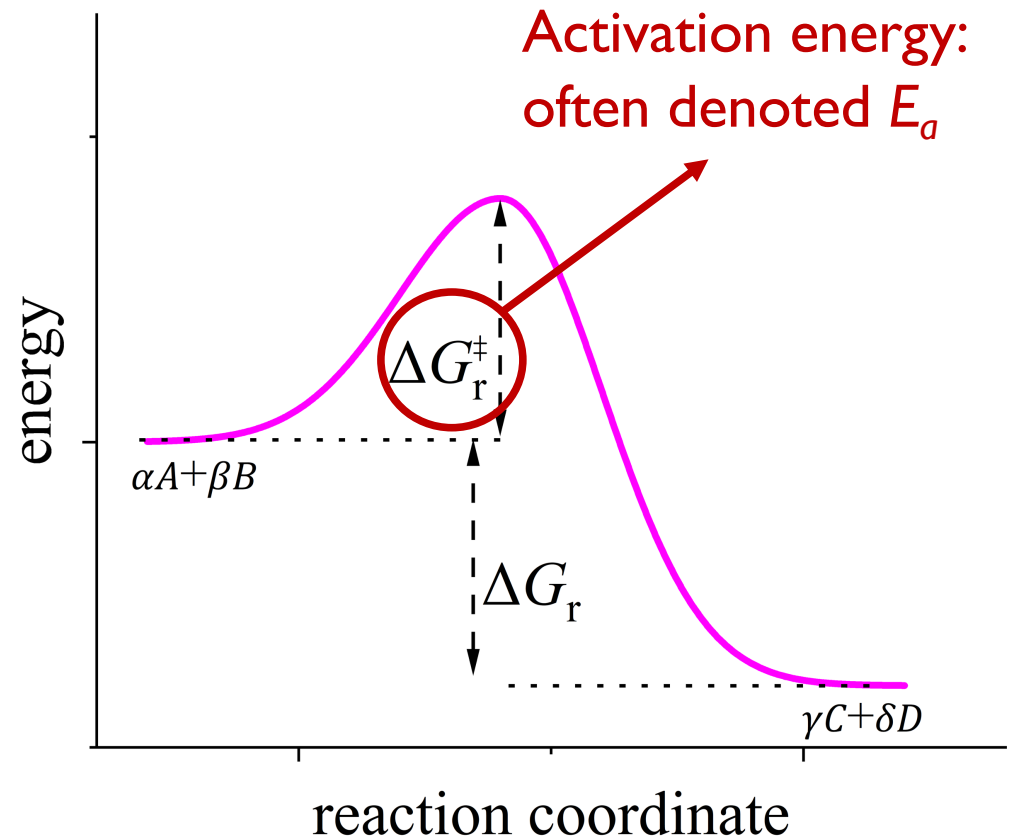
First order in $[\text{NH}_4^+]$

First order in $[\text{NO}_2^-]$

- The overall reaction order can be found by adding the exponents on the reactants in the rate law.
This reaction is *second-order overall*.

Temperature and rate

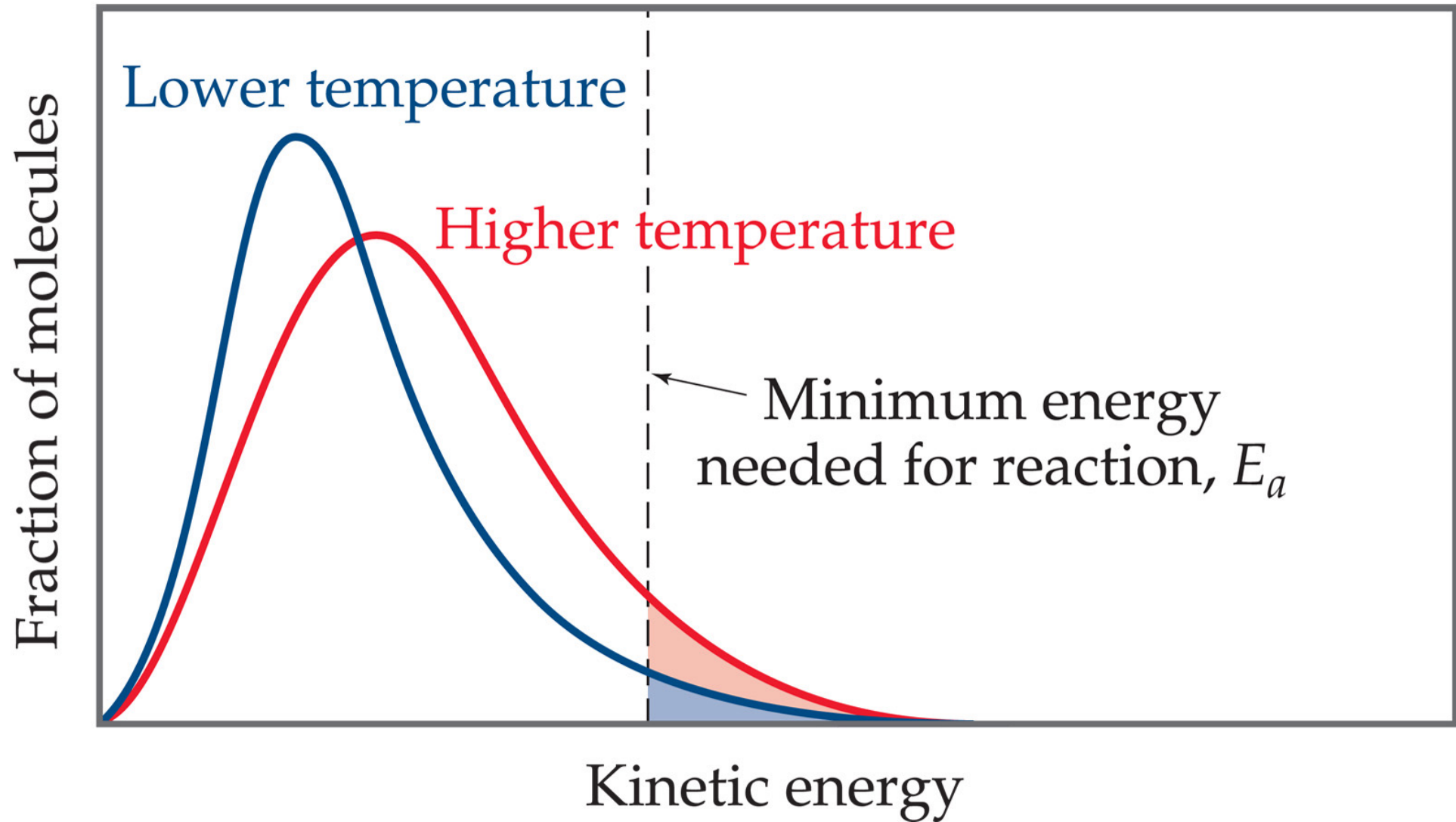
Every reaction passes through a state with high energy (but the lowest energy possible on the path towards product), called the **transition state or activated complex(TS)**.



The Collision Model

- In a chemical reaction, bonds are broken and new bonds are formed.
- Molecules can only react if they collide with each other.

Maxwell-Boltzmann distribution



The Arrhenius equation

The fraction, f , of molecules with kinetic energy higher than E_a can be shown to be $f = \exp(-E_a/RT)$ (R is the gas constant and T is the temperature in K)

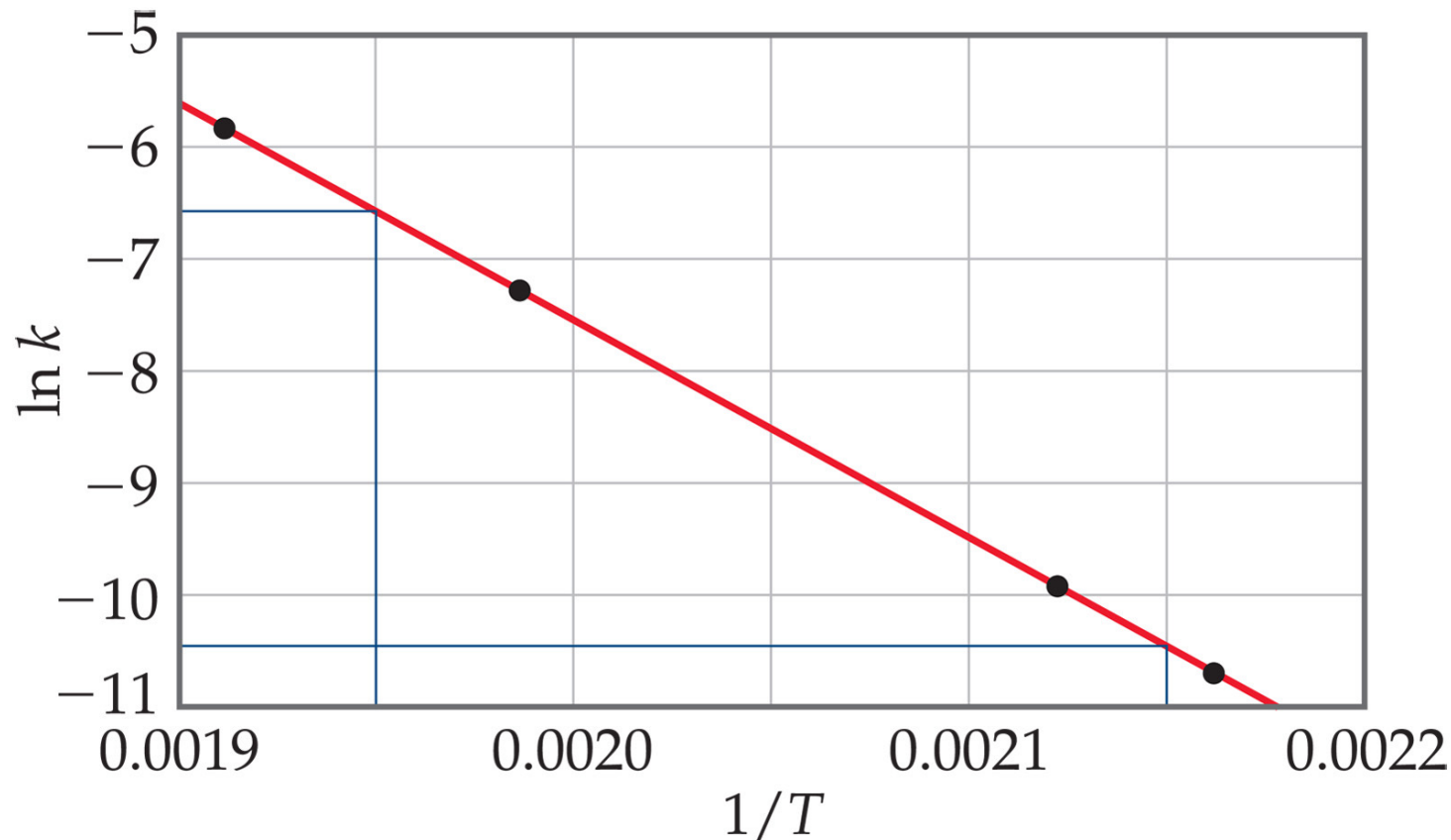
Swedish chemist Svante Arrhenius realized the relationship between k and E_a :

$$k = A \exp(-E_a/RT)$$

where A is the frequency factor, a number that represents the likelihood that collisions would occur with the proper orientation for reaction.

The Arrhenius equation

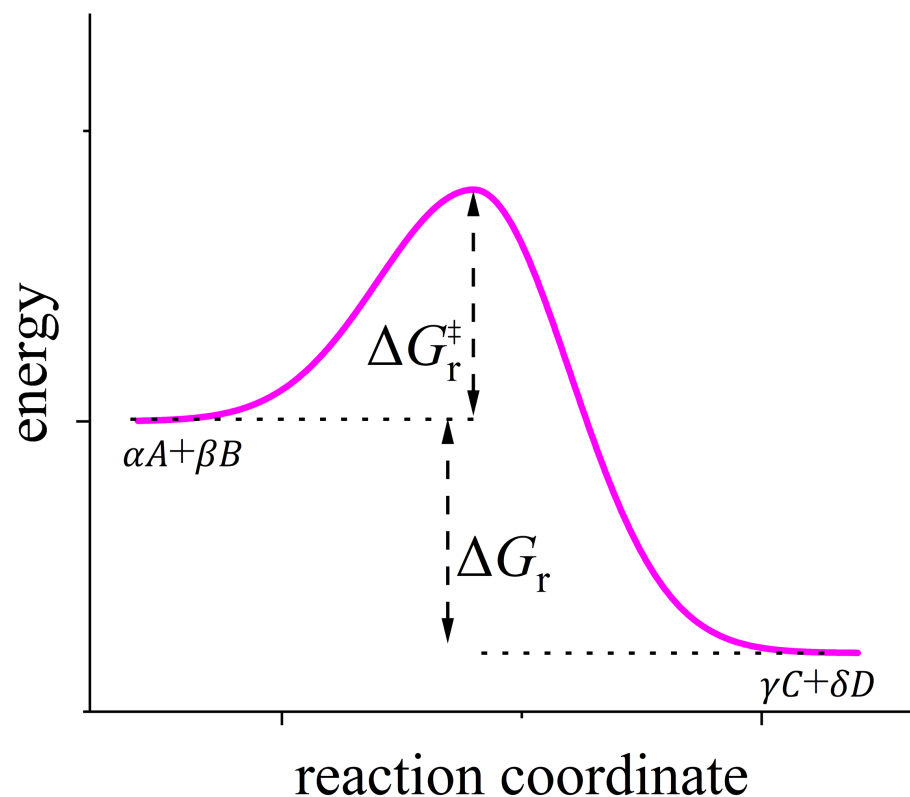
Taking the logarithm yields: $\ln(k) = -E_a/RT + \ln(A)$



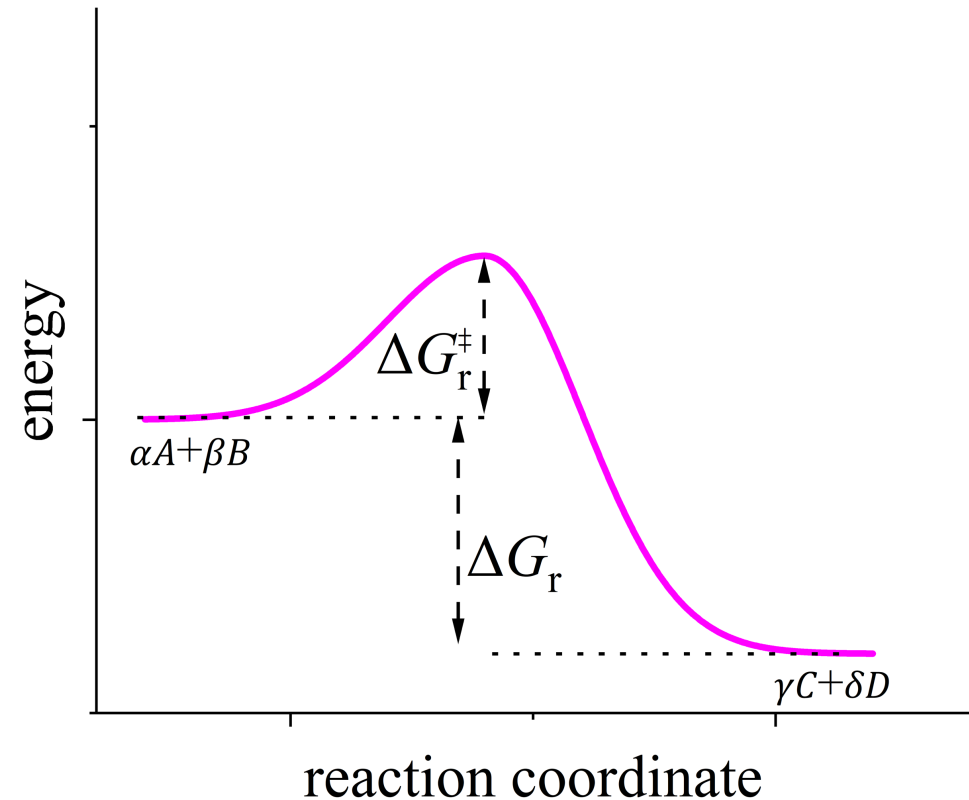
When k is determined experimentally at several temperatures, E_a can be calculated from the slope of a plot of $\ln k$ vs. $1/T$.

Activation energy & catalysis

Catalysis: lowering of the barrier for reaction (energy of TS) leaves more molecules to have sufficient energy to pass to products



without catalysis



with catalysis

Integrated rate laws: First order

Consider a simple reaction: $A \longrightarrow B$
with the *first order* rate law $r = k [A]$

Rearranging and integration between $t = 0$ and $t = t$ gives

$$-\frac{d[A]}{dt} = k[A] \Leftrightarrow \frac{1}{[A]} d[A] = -k dt$$

integration between $t = 0$ and $t = t$ gives

$$\int_{[A]_0}^{[A]_t} \frac{1}{[A]} d[A] = -k \int_0^t dt$$



$$[A]_t = [A]_0 \exp(-kt)$$

Or alternatively:

$$\ln[A]_t = -kt + \ln[A]_0$$

y = ax + b

Integrated rate laws: Second order

Consider a simple reaction: $A \longrightarrow B$
with the **second order** rate law $r = k [A]^2$

Rearranging and integration between $t = 0$ and $t = t$ gives

$$-\frac{d[A]}{dt} = k[A]^2 \Leftrightarrow \frac{1}{[A]^2} d[A] = -k dt$$

integration between $t = 0$ and $t = t$ gives

$$\int_{[A]_0}^{[A]_t} \frac{1}{[A]^2} d[A] = -k \int_0^t dt$$


$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

$y = ax + b$

Exercise: Determine reaction order and rate constant

The decomposition of NO_2 at 300 °C is described by the equation



and yields these data:

Time (s)	$p(\text{NO}_2)$ (bar)
0.0	0.01000
50.0	0.00787
100.0	0.00649
200.0	0.00481
300.0	0.00380

First order:

$$\ln[A]_t = -kt + \ln[A]_0$$

Second order:

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

Determining reaction order

Half-life for first order process

From the integrated rate law: $\ln[A]_t = -kt + \ln[A]_0$

it follows that $\ln \frac{[A]_t}{[A]_0} = -kt$

After one half-life:

$$\ln \frac{\frac{1}{2}[A]_0}{[A]_0} = -kt_{\frac{1}{2}}$$

$\Leftrightarrow$

$$t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

NOTE: For a first-order process, the half-life *does not* depend on $[A]_0$.

Half-life for second order process

From the integrated rate law: $\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$

we get after one half-life:

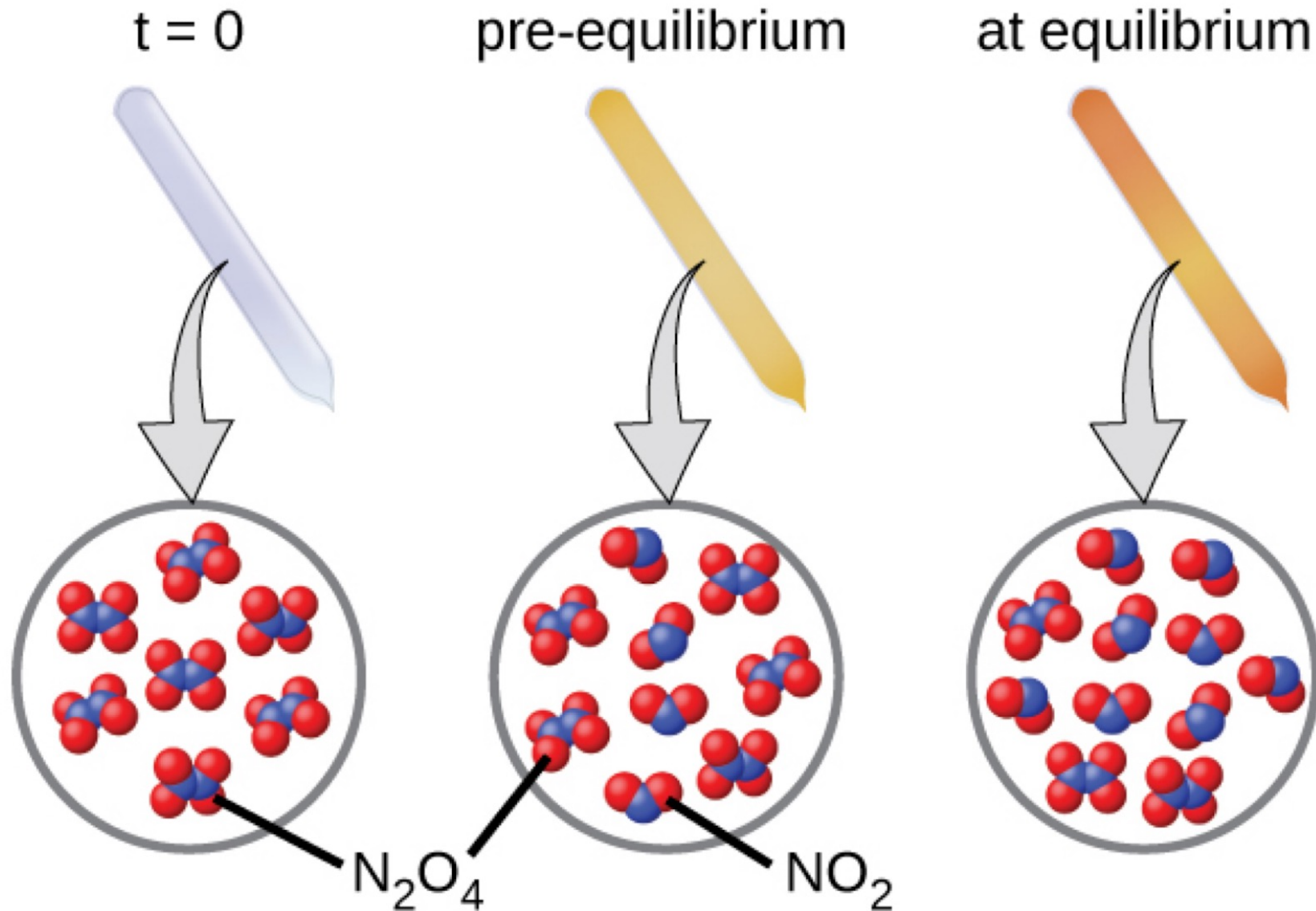
$$\frac{1}{1/2[A]_0} = kt_{\frac{1}{2}} + \frac{1}{[A]_0}$$

$\Leftrightarrow$

$$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$$

NOTE: For a second-order process, the half-life *does* depend on $[A]_0$.

The concept of equilibrium

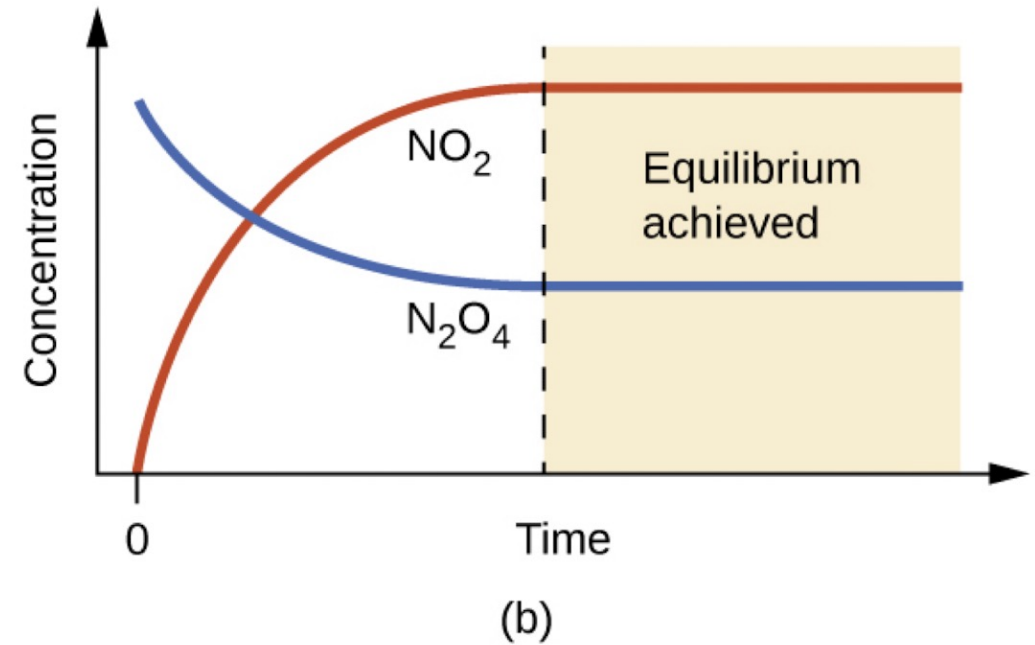


The concept of equilibrium

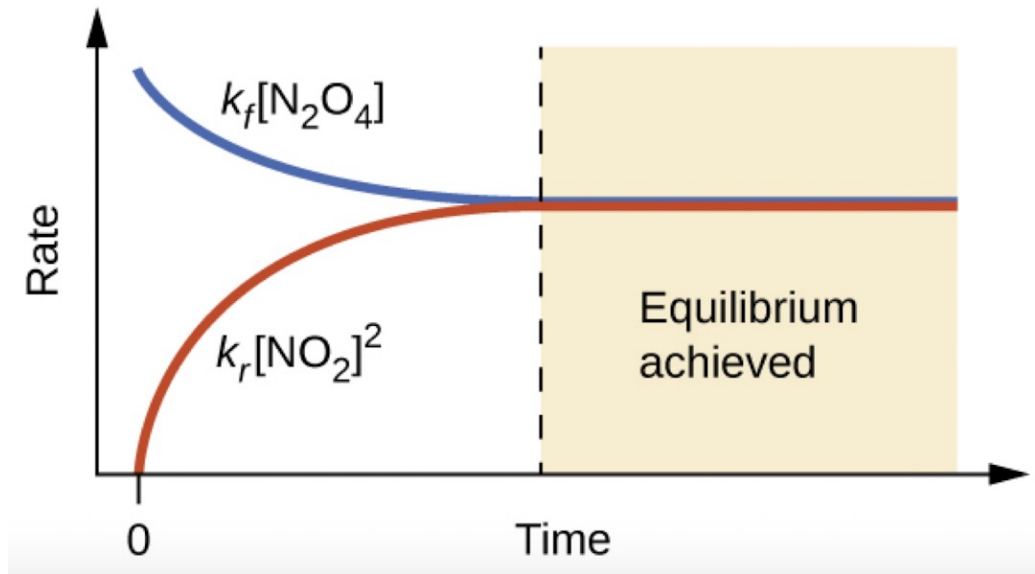
A *forward* reaction



and the *reverse*



At this time, **equilibrium** is obtained:



From earlier (Ch. 16)

Gibbs free energy change:

$$\Delta G = \Delta H - T \Delta S$$

$\Delta G < 0$ for a spontaneous process

$\Delta G > 0$ for a nonspontaneous process

$\Delta G = 0$ for a process at equilibrium

Gibbs free energy under non-standard conditions (Ch. 16)

The relationship between ΔG° and ΔG is given by:

$$\Delta G = \Delta G^\circ + RT \ln(Q)$$

where Q is the **reaction quotient**.

For a general reaction,



Q is given by

$$Q = \frac{[C]^\gamma [D]^\delta}{[A]^\alpha [B]^\beta}$$

At equilibrium (from Ch. 16)

$\Delta G = 0$, and Q becomes the *equilibrium constant*, K :

$$0 = \Delta G^\circ + RT \ln(K)$$

Which allows the equilibrium constant to be determined from knowing ΔG° :

$$K = \exp(-\Delta G^\circ/RT)$$

K	$\ln(K)$	ΔG°	Comments
> 1	Positive	Negative	Products favored over reactants
$= 1$	0	0	Products and reactants equally favored
< 1	Negative	Positive	Reactants are favored over products

Rules for K expressions

1. Concentrations in solution are given in M, in gas phase typically in atm or bar.
2. The concentration of pure solids, pure liquids (heterogeneous equilibria), and solvents do not appear in the expressions.

EXAMPLE: *Solubility product*

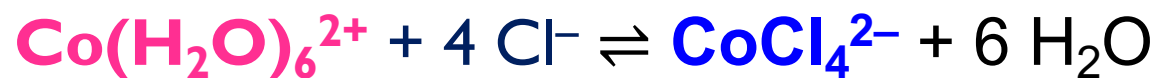
3. In quoting a value for the equilibrium constant, we must specify the balanced equation and the temperature.

Le Châtelier's principle

*If an **equilibrium** is disturbed, the position of equilibrium shifts to counteract the change to reestablish an equilibrium.*

If a chemical reaction is at equilibrium and experiences a change in pressure, temperature, or concentration of products or reactants, the equilibrium shifts in the opposite direction to offset the change.

EXAMPLE:



Example exercise:

Consider again the reaction $2 \text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$

- ▶ Write down the reaction quotient.
- ▶ A 1 L container is filled with 1 atm of pure NO_2 . Given that the equilibrium constant is $K = 6.1 \text{ atm}^{-1}$ at 25°C , calculate $p(\text{NO}_2)$ and $p(\text{N}_2\text{O}_4)$ at equilibrium.
- ▶ The volume of the contained is reduced to 0.5 L using a piston. Calculate the reaction quotient before equilibrium has been established. In which direction will the equilibrium shift?
- ▶ The reaction's ΔH° is -580 kJ mol^{-1} . Using Le Chatelier's principle, would cooling (to e.g. -78°C) increase or decrease the amount of NO_2 ? Can you explain that using the concept of entropy?